Nguyen Minh Thuan

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Education

International University – VNU-HCM Ho Chi Minh City, Vietnam

Bachelor of Computer Science and Engineering, Cumulative GPA: 3.0 / 4.0 *Expected Graduation: 2026*

- **Relevant Coursework**: Artificial Intelligence, Machine Learning, Data Mining, Database Management, Linear Algebra, Probability & Statistics, Software Engineering, Computer Networks.

Technical Skills

Core AI & ML: PyTorch, Scikit-learn, Pandas, NumPy

GenAI & Agents: LangChain, LangGraph, Milvus, ChromaDB

Backend & DevOps: FastAPI, Docker, PostgreSQL, MySQL, Git, CI/CD

Languages: Python, SQL, C++, JavaScript

Experience

Information Technology Park (ITP), VNU-HCM Feb 2026 – Jul 2026

Software Engineer Intern *Ho Chi Minh City, Vietnam*

- Engineered REST APIs for VNU-HCM MOOC infrastructure serving 10,000+ learners, sustaining sub-120ms response latency under 500-concurrent-user Apache JMeter stress testing.
- Constructed an end-to-end data pipeline using Python and MySQL, ingesting and normalizing 152,714+ real-time xAPI interaction logs.
- Decoupled iframe content delivery from LTI 1.3 protocol handshakes, reducing Time-To-First-Byte by 312× (1,904.7ms to 6.1ms) and page load times by 4.9×.

HIPTECH Co., LTD Jun 2025 – Jan 2026

AI Engineer Intern *Ho Chi Minh City, Vietnam*

- Developed backend inference and ingestion services for 2 internal GenAI systems (WARP and Corsair Scanner).
- Containerized application inference stacks using Docker, maintaining 99.5% system availability across upstream integration sources.
- Implemented PostgreSQL state management and persistence layers for vector and tabular datasets.

Projects

Traceable Voice Copilot for Digital Cockpit Jul 2026 – Aug 2026

Python, FastAPI, Kotlin, ChromaDB, Docker | Link

- Built an in-vehicle Voice Copilot on Android Automotive OS using Agentic RAG to query vehicle manuals.
- Engineered a hybrid BM25 + dense ChromaDB retrieval pipeline using Reciprocal Rank Fusion (RRF) and Cross-Encoder reranking.
- Raised retrieval accuracy across 84 vehicle manuals from 76% to 100% by implementing a RapidOCR fallback parser for embedded diagrams and tables.
- Engineered a FastAPI Intent Router (<5ms latency) with Citation Gate verification, and provisioned AWS deployment pipelines via Ansible and Terraform.

Capstone Project: Learning Telemetry & Analytics Platform Jan 2026 – Jun 2026

Python, Playwright, MySQL, WordPress | Link

- Engineered an xAPI telemetry capture handler to record student learning milestones, video drop-offs, and assessment scores into custom MySQL schemas.
- Developed an automated benchmarking harness using Playwright and Chrome DevTools Protocol (CDP) to measure TTFB, FCP, and Video Startup Time.
- Statistically validated content delivery latency improvements across 5 sessions using the non-parametric Mann-Whitney U test ($U = 0, p < 0.01$).

- Exposed aggregated learning metrics via REST endpoints, sustaining sub-120ms response times for real-time dashboard consumption.

WARP: Multi-Agent Recommendation System Nov 2025 – Jan 2026

Python, LangChain, LangGraph, Milvus, Docker

- Built a semantic search pipeline across 50,000+ catalog items using E5 embeddings and Milvus, achieving sub-250ms search latency.
- Implemented stateful multi-agent workflows using LangGraph and FastAPI to execute multi-criteria compatibility validation.

Corsair Scanner: Automated Hardware Compatibility Engine Sep 2025 – Nov 2025

Python, PostgreSQL, FastAPI

- Developed a hybrid SQL and vector verification pipeline across 50,000+ hardware SKUs with sub-500ms query latency.
- Implemented anomaly detection and schema validation layers to enforce data integrity across incoming ingestion pipelines.

Awards & Competitions

Top 13 / 300+ Teams – FPT Automotive Hackathon 2026 Ho Chi Minh City, Vietnam

Team Leader & AI Engineer (Team Wheelchair) *Jul 2026 – Aug 2026*

- Led a 5-member team (Wheelchair) through a 4-week sprint to build and pitch the in-vehicle Agentic AI copilot, placing Top 13 out of 300+ national teams.

References

MSc. Le Thanh Son | *Thesis Supervisor, Lecturer*
International University – VNU-HCM
Email: ltson@hcmiu.edu.vn

MSc. Nguyen Huu Viet Long | *Technical Mentor, Staff*
Information Technology Park (ITP) – VNU-HCM

Mr. Tran Thien Khiem | *Chief Technology Officer (CTO)*
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